

Fertilizer in Two Greenhouses

Unit 1 · Topic 1.13 · THE PROBLEM

Suppose a grower has 24 similar tomato plants, 12 in a sunny greenhouse and 12 in a shaded greenhouse. The response is fruit yield in kilograms per plant.

Design A: Completely randomize all 24 plants; assign 12 to fertilizer and 12 to no fertilizer, ignoring greenhouse.

Design B: Treat greenhouse as a block; within each greenhouse, randomly assign 6 plants to fertilizer and 6 to no fertilizer.

Pilot means from plants grown under the same conditions are shown.

Mean fruit yield (kg per plant)

Greenhouse	Fertilizer	None
Sunny	7.2	6.0
Shaded	4.0	2.8

FIRST TAKE — one sentence before you work the parts

Which design would you use to detect the fertilizer effect most clearly? Name one detail behind your choice.

- DOK 1** (a) Name the blocking variable and the response variable, with units.
- DOK 2** (b) For the sunny greenhouse, find fertilizer minus no fertilizer. For the fertilized plants, find sunny minus shaded. Compare the two differences.
- DOK 3** (c) Which design should the grower use? Use the two differences to explain your choice and how chance imbalance could affect Design A. If Design B gives a statistically significant difference, state a causal conclusion limited to the plants and conditions studied.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.